

TONGSEOK LIM

◇ <https://tlim0213.github.io> ◇ lim336 at purdue dot edu ◇ September 25, 2026

ACADEMIC CAREER

- **Purdue University, West Lafayette, IN, USA** Aug 2020 – present
Assistant Professor in Quantitative Methods, Mitch Daniels School of Business
Homepage: <https://tlim0213.github.io>
- **Fields Institute, Toronto, Canada** Mar – Jul 2019 & Mar – Jul 2020
Fields Research Fellow
- **ShanghaiTech University, Shanghai, China** Sep 2018 – Jun 2020
Assistant Professor of Mathematics
- **University of Oxford, Oxford, UK** Oct 2016 – Aug 2018
Postdoctoral Research Assistant
- **Technische Universität Wien, Vienna, Austria** May – Sep 2016
Postdoctoral Research Assistant
- **University of British Columbia, Vancouver, Canada** Sep 2011 – May 2016
PhD in Mathematics
- **Citizenship: Republic of Korea | Permanent Residency: United States**

RESEARCH INTERESTS

Optimization and mathematical foundations of data science; optimal transport, Schrödinger bridges, and Gibbs variational methods; data integration, statistical learning, and decision-making under uncertainty.

PUBLICATIONS

- **On the structure of optimal martingale transport in higher dimensions.** *PhD thesis*. [Link](#)
- with N. Ghoussoub and Y-H. Kim. **Structure of optimal martingale transport in general dimensions.** *Annals of Probability*, 2019. [Journal](#)
- with M. Beiglböck and J. Oblój. **Dual attainment for the martingale transport problem.** *Bernoulli*, 2019. [Journal](#)
- **Optimal martingale transport between radially symmetric marginals in general dimensions.** *Stochastic Processes and their Applications*, 2020. [Journal](#)
- with N. Ghoussoub and Y-H. Kim. **Optimal Brownian stopping when the source and target are radially symmetric distributions.** *SIAM Journal on Control and Optimization*, 2020. [Journal](#)
- with S. Eckstein, G. Guo and J. Oblój. **Robust pricing and hedging of options on multiple assets and its numerics.** *SIAM Journal on Financial Mathematics*, 2021. [Journal](#) [arXiv](#)
- with K. Kang, H. Kim and G. Seo. **Uniqueness and characterization of local minimizers for the interaction energy with mildly repulsive potentials.** *Calculus of Variations and Partial Differential Equations*, 2021. [Journal](#) [arXiv](#)
- with R.J. McCann. **Isodiametry, variance, and regular simplices from particle interactions.** *Archive for Rational Mechanics and Analysis*, 2021. [Journal](#) [arXiv](#)
- with R.J. McCann. **On Fejes Tóth's conjectured maximizer for the sum of angles between lines.** *Applied Mathematics and Optimization*, 2021. [Journal](#) [arXiv](#)

- with R.J. McCann. **Geometrical bounds for the variance and recentered moments.** *Mathematics of Operations Research*, 2022. [Journal](#) [arXiv](#)
- with R.J. McCann. **Maximizing expected powers of the angle between pairs of points in projective space.** *Probability Theory and Related Fields*, 2022. [Journal](#) [arXiv](#)
- with R.J. McCann. **On the cardinality of sets in $\mathbb{R}^d$ obeying a slightly obtuse angle bound.** *SIAM Journal on Discrete Mathematics*, 2022. [Journal](#) [arXiv](#)
- with C. Davies and R.J. McCann. **Classifying minimum energy states for interacting particles: Spherical shells.** *SIAM Journal on Applied Mathematics*, 2022. [Journal](#) [arXiv](#)
- with C. Davies and R.J. McCann. **Classifying minimum energy states for interacting particles: Regular simplices.** *Communications in Mathematical Physics*, 2023. [Journal](#) [GitHub](#) [arXiv](#)
- **Maximal monotonicity and cyclic involutivity of multiconjugate convex functions.** *SIAM Journal on Optimization*, 2023. [Journal](#) [GitHub](#) [arXiv](#)
- **Geometry of vectorial martingale optimal transportations and duality.** *Mathematical Programming* (series A), 2024. [Journal](#) [GitHub](#) [arXiv](#)
- with E. Koo. **Node classification in networks via simplicial interactions.** *IEEE Transactions on Neural Networks and Learning Systems*, 2025. [Journal](#) [GitHub](#) [arXiv](#)
- with E. Koo. **Node classification via simplicial interaction with augmented maximal clique selection.** *Neurocomputing*, 2025. [Journal](#) [GitHub](#) [arXiv](#)
- with J. Hiew, B. Pass and M. Souza. **Data denoising with self consistency, variance maximization and the Kantorovich dominance.** *SIAM Journal on Mathematics of Data Science*, 2026. [Journal](#) [arXiv](#)
- with E. Koo and J. Sohn. **Statistical matching via Schrödinger bridge beyond conditional independence.** *The 40th Conference on Neural Information Processing Systems*, to appear. [GitHub](#) [arXiv](#)

PAPERS UNDER REVIEW

- with K. Nam and J. Sohn. **Monotone curve estimation via convex duality.** [GitHub](#) [arXiv](#)
- with C. Che and Y. Sun. **Dual attainment in multi-period multi-asset martingale optimal transport and its computation.** [arXiv](#)
- **Stable allocations, Unstable payment paths: The Shapley value and voluntary coalition formation.** [GitHub](#) [arXiv](#)
- with W. Ha and K. Nam. **Statistical guarantees for multimodal contrastive learning via pointwise mutual information estimation.**

WORK IN PROGRESS

- with W. Ha and K. Nam. **Learning implicit multivariate interactions via Gibbs relaxation.**
- with J. Hiew, B. Pass and M. Souza. **Dimension reduction in martingale optimal transport: geometry and robust option pricing.** [GitHub](#) [arXiv](#)
- with C. Che, P. Das and Y. Sun. **Observable-augmented martingale optimal transport for robust pricing and hedging of multi-asset exotics.**
- **Optimal exercise decision of American options under model uncertainty.** [GitHub](#) [arXiv](#)
- **Axiomatic and probabilistic foundations for the Hodge-theoretic Shapley value.** [GitHub](#)
- with D. Jeong and E. Shin. **Robust intervention in networks.** [GitHub](#) [arXiv](#)
- with D. Jeong and E. Shin. **Value allocation in coalition networks.** preliminary version on [arXiv](#)
- with J. Kil. **From pairs to teams: sparse flexibility with complementary resources.**

INVITED TALKS & LECTURES

- Aug 2026. *9th International Conference on Econometrics and Statistics*. Ryukoku University, Japan. [Link](#)
- Jul 2026. *2026 Asian Quantitative Finance Conference*. Lahan Hotel Pohang, Korea. [Link](#)
- Aug 2025. *Workshop on Quantum Analysis & Geometry*. Lakai Sandpine Resort, Korea. [Link](#)
- Jul 2025. *Statistics seminar*. The Chinese University of Hong Kong.
- May 2025. *Workshop on Mathematical Finance at Seoul*. Seoul National University, Korea. [Link](#)
- Jul 2024. *Applied Mathematics seminar*. Nanyang Technological University, Singapore.
- Jun 2024. *Optimal Transport, Stochastic Analysis and Machine Learning*. KAIST, Korea. [Link](#)
- May 2024. *Workshop on Mathematical Finance at Jeju*. Jeju Island, Korea. [Link](#)
- Mar 2024. *Financial and Actuarial Mathematics seminar*. University of Michigan, USA. [Link](#)
- May 2023. *Conference on Approximation Theory and Beyond*. Vanderbilt University, USA. [Link](#)
- Apr 2023. *Midwest International Trade & Theory Conference*. University of Tennessee, USA. [Link](#)
- Mar 2023. *Finance seminar*. College of Business, Stony Brook University, USA.
- Mar 2023. *Eighth annual conference on network science and economics*. Virginia Tech, USA. [Link](#)
- Feb 2023. *Quantitative Methods seminar*. School of Business, Purdue University, USA.
- Feb 2023. *Analysis & PDE seminar*. Department of Mathematics, Purdue University, USA.
- Jan 2023. *2023 Joint Mathematics Meetings (JMM 2023)*. Boston, USA. [Link](#)
- Jul 2022. *AI seminar*. Center for AI and Natural Sciences, KIAS, Korea. [Link](#)
- Jun 2022. *Colloquium seminar*. Kyungpook National University, Korea.
- May 2022. *Colloquium seminars I & II*. Pusan National University, Korea.
- May 2022. *AI seminar*. Center for AI and Natural Sciences, KIAS, Korea. [Link](#)
- Mar 2022. *Stochastic Mass Transports*. Banff International Research Station. [Link](#) [Video](#)
- Dec 2021. *Analysis & PDE seminar I & II*. Yonsei University, Korea.
- Oct 2021. *Game theory seminar*. College of business, KAIST, Korea.
- Sep 2021. *Mathematics & Economics joint seminar*. Washington University in St. Louis, USA. [Link](#)
- Dec 2020. *Canadian Mathematical Society Winter Meeting*. online. [Link](#) [Video](#)
- Feb 2020. *Colloquium*. Krannert School of Management, Purdue University, USA.
- Oct 2019. *Korean Mathematical Society (KMS) Annual Meeting*. Hong-Ik University, Seoul, Korea.
- Oct 2019. *Analysis & PDE Seminar*. Georgia Institute of Technology, USA. [Link](#)
- Apr 2019. *Analysis & Applied Math Seminar*. University of Toronto, Canada.
- Mar 2019. *Colloquium Seminar*. Pusan National University, Korea.
- Aug 2018. *Mathematics Seminar*. Seoul National University, Korea.
- Jun 2018. *Mathematical Finance Internal Seminar*. University of Oxford, UK.
- May 2018. *CMO Workshop 18w5080: Stochastic Analysis and its Applications*. Mexico. [Link](#) [Video](#)
- May 2018. *Mathematics Seminar*. University of Bath, UK.
- Apr 2018. *Mathematics Seminar*. University of Central Florida, USA.
- Dec 2017. *Vienna Seminar in Mathematical Finance and Probability*. University of Vienna, Austria.
- Dec 2017. *Mathematics Seminar*. Dublin City University, Ireland.
- Sep 2017. *Martingale Optimal Transport (and Friends)*. University of Oxford, UK. [Link](#)

- Jul 2017. *Stochastic Processes and their Applications (SPA) 2017*. Moscow, Russia. [Link](#)
- May 2017. *Oxford-Princeton Workshop on Financial Mathematics & Stochastic Analysis*. UK. [Link](#)
- Jan 2017. *Mathematics Seminar*. Strasbourg University, France.
- Jan 2017. *Advances in Financial Mathematics*. Paris, France. [Link](#)
- Jan 2017. *Mathematics Seminar*. Ecole Polytechnique, Palaiseau, France.
- Jan 2017. *Paris Bachelier Seminar*. Paris, France. [Link](#)
- Nov 2016. *Mathematical Finance Internal Seminar*. University of Oxford, UK. [Link](#)
- Nov 2016. *Model Uncertainty & Robust Finance*. University of Milan, Italy.
- Jul 2016. *Mathematics Seminar*. Scuola Normale Superiore, Pisa, Italy.
- Jun 2016. *Mathematics Seminar*. Yonsei University, Seoul, Korea.
- Apr 2016. *Optimal Transportation, Equilibrium, and Applications to Economics*. NYU, USA. [Link](#)
- Mar 2016. *Skorokhod embeddings, Martingale Optimal Transport and their applications*. UK. [Link](#)
- Feb 2016. *Vienna Seminar in Mathematical Finance and Probability*. TU Wien, Austria. [Link](#)
- Sep 2015. *Mathematics Seminar*. Ecole Polytechnique, Palaiseau, France.
- Sep 2015. *European Summer School In Financial Mathematics*. Le Mans, France. [Link](#)
- Aug 2015. *KIAS CMC minischool on Analysis, Geometry, and Optimal Transport*. KIAS, Korea. [Link](#)

RESEARCH STUDENTS AND ALUMNI

PhD supervision

- Marcelo Cruz de Souza (Defended in July 2024)
- Joshua Zoen-Git Hiew (Defended in 2025 at Univ. of Alberta. supervised jointly with Brendan Pass)
- Junhyeok Kil (Started in January 2026)

MSc supervision

- Cameron Davies (currently PhD student at Univ. of Toronto, supervised jointly with Robert McCann)

GRANT PROPOSALS

NSF CAREER: “Gibbs Variational Methods for Data Fusion, Imputation, and Interaction Learning.”
Principal Investigator, \$599,465. Submitted July 2026; under review.

TEACHING EXPERIENCE

Daniels School of Business, Purdue University	2020 – present
<i>Assistant Professor in Quantitative Methods</i>	<i>West Lafayette, IN, USA</i>
· MGMT 30500–001,002,003 : Business Statistics	Spring 2027
· MGMT 30500–002,003 : Business Statistics	Spring 2026
· MGMT 69000 : Optimal Transport and Unsupervised Learning	Spring 2025
· MGMT 47400–001,002 : Predictive Analytics	Fall 2024
· MGMT 47400–001 : Predictive Analytics	Spring 2024
· MGMT 47400–001,002 : Predictive Analytics	Fall 2023
· MGMT 47400–001 : Predictive Analytics	Spring 2023
· MGMT 47400–001,002 : Predictive Analytics	Fall 2022
· MGMT 69000 : Advanced Problems in Management – Optimal Transport Problems	Spring 2022
· MGMT 47400–001 : Predictive Analytics	Spring 2022
· MGMT 47400–001,002 : Predictive Analytics	Fall 2021

- MGMT 47400–001,002 : Predictive Analytics Spring 2021

- ShanghaiTech University**
Assistant Professor of Mathematics

2018 – 2020
Shanghai, China
- Stochastic Processes (for 3rd & 4th grades) 2019
- Linear Algebra (for 2nd grades) 2018

- University of Oxford**
Class Tutor

2018
Oxford, UK
- Class Tutor for B8.3 – Mathematical Models of Financial Derivatives 2018

- University of British Columbia**
Teaching Assistant and Instructor

2011 – 2016
Vancouver, BC, Canada
- Instructor for MATH 101 – Integral Calculus with Applications to Physics and Engineering 2016
- Instructor for Workshops – Direct student groups towards solving math problems 2015
- TA for MATH 121 – Honours Integral Calculus 2015
- TA for MATH 441 – Mathematical Modeling: Discrete Optimization Problems 2014
- TA for MATH 424 – Classical Differential Geometry 2012 – 2013

HONOURS & AWARDS

- Fields Research Fellowship. 2019 & 2020. University of Toronto, Canada.
- Four Year Doctoral Fellowship (4YF) recipient. 2011 – 2015. University of British Columbia, Canada.
- Kwanjeong Scholarship recipient. 2011 – 2016. Kwanjeong Educational Foundation, Seoul, Korea. [Link](#)
- Seoul Scholarship recipient. 2008 – 2010. Seoul Metropolitan Government, Korea.